

From Scalar Reward to Cellular Credit

Ognjen Čadovski

Doctoral student, Faculty of Mathematics, University of Belgrade

Supervisor: Aleksandar Takači

ognjencadovski@gmail.com

September 2026

Abstract

Learning from a single outcome poses two problems. A learner must identify which population pattern produced the outcome and determine how each cell should change. Francioni et al. found dendritic signals that depended on each neuron’s role in a brain computer interface task [Francioni et al., 2026]. We asked whether feedback and population selection in ART could explain these signals without a separate error for every neuron. We tested this idea in a sequence of simulations based on pART and SMART. The models learned useful behavior, but the detailed pattern across neurons came mainly from population patterns already available. The complete model also failed its main test because its early dendritic signal did not predict later changes in weights and cell activity. We therefore tested a standard perturbation rule in which each cell retains a local eligibility trace of its recent change. This rule learned new cellular directions from a scalar reward. A common benchmark compared ART, local eligibility, their direct combination, a random control, and an oracle given the correct directions. ART selected useful existing patterns. Eligibility learned a new pattern across cells. Their combination improved behavior and topology, but its dendritic signal had the wrong sign. Only the oracle passed every criterion. Population selection and cellular learning are therefore distinct operations. They may work together in the brain, but their direct combination does not yet explain the biological signal.

Keywords credit assignment, Adaptive Resonance Theory, dendritic learning, eligibility traces, scalar reward, brain computer interface

1 Introduction

Many theories of learning begin with an error. They compare a desired outcome with the outcome that occurred, then use the difference to determine how the network should change. This approach is powerful because it connects behavior to adjustments at individual units or synapses. Its biological implementation is uncertain. The brain must obtain appropriately structured teaching information and deliver it to the cells that can improve the outcome.

A different tradition begins with population representations. Adaptive Resonance Theory (ART) describes how neural systems learn stable categories, form expectations through feedback, and search when an expectation does not match the current input [Carpenter and Grossberg, 1987]. Predictive ART (pART) extends this view to consequences that occur later. It asks which active representation predicted the consequence and should therefore be reinforced [Grossberg, 2018]. It does not describe learning mainly as the reduction of a separate error at every neuron.

The original motivation for this study came from findings by Francioni et al. that we encountered while studying ART. Their closed loop brain computer interface (BCI) task revealed dendritic

signals whose signs depended on the causal role of each neuron [Francioni et al., 2026]. The signals contained information about reward and error, predicted later changes in neural activity, and were necessary for learning under the reported intervention. They can be understood as error or target signals for individual neurons. Yet they also involve feedback to dendrites. This raised a simple question. Could a cellular result that appears to be driven by error instead arise from the kind of population learning described by ART?

We turned this into a specific computational question. Could pART select a causal representation and learn a feedback pattern that reproduced the sequence reported by Francioni et al.? In particular, could an early dendritic signal predict later changes in each neuron even though the learner received no separate error for each cell? pART was a reasonable starting point because it addresses delayed consequences and the selection of causal representations. ART provides learned expectations, while SMART provides a model of apical feedback, matching, timing, and local plasticity [Grossberg and Versace, 2008]. Together, these ideas suggested a possible path from a delayed scalar outcome to changes in individual cells. The central issue was whether that path contained enough information.

That possible information gap is the central problem examined here. An outcome can arrive after the action that caused it, creating a temporal credit problem. Many cells can also be active at once while contributing differently to the outcome, creating a structural credit problem. A single scalar reward reports whether the joint action helped. On its own, it does not identify the contribution of each cell.

We use *representation credit* to mean choosing and reinforcing a useful population pattern that is already available. We use *cellular credit* for the more detailed problem of deciding the direction and size of change at each cell. We call the resulting arrangement across cells the *topology* of the learned signal. A system can select a useful representation without learning a new topology.

This distinction is especially important in a BCI task in which nearby cortical cells have opposite causal roles. One possible learning mechanism computes or supplies a different teaching value for every neuron. Another broadcasts one scalar outcome and combines it with information stored locally at each cell. A third selects a useful population pattern that already exists. These mechanisms can produce similar behavior, but they solve different problems.

We studied this question in stages, with each experiment addressing a limitation found in the previous one. We first asked whether selective feedback could route an opponent pattern that was already built into the model. We then hid the roles of the cells and required the model to select from random population patterns. Next, the model learned its own feedback expectation instead of receiving a copied pattern. We tested a simpler local plasticity mechanism by itself before adding it to the complete model. We then traced where the detailed pattern across neurons had come from. Another experiment asked whether a local memory of recent exploration could supply the missing information. The final benchmark placed ART selection, local eligibility, and their direct combination in the same task. It evaluated behavior, the learned pattern across cells, and the dendritic signal separately.

This order is important. Successful behavior alone does not reveal whether the learner selected an existing representation or discovered how individual cells should change.

The contribution is therefore not a new general learning algorithm. It is a controlled comparison of two levels of explanation. The experiments separate the selection of a population pattern from learning the direction of change at each cell. They also test whether an early dendritic signal predicts later change, and whether directly combining the two operations explains the missing step. All numerical results and figures come from the archived repository record. Claims about published theory and biology are supported separately by external citations.

2 Related work

2.1 Error, target, and dendritic accounts

Backpropagation solves credit assignment in artificial networks by using the derivative of an objective to calculate coordinated weight changes. Models that aim to be more biologically plausible replace exact backpropagation with feedback that can be processed locally. Models with several dendritic compartments use basal dendrites for incoming activity and apical dendrites for teaching or target signals [Guerguiev et al., 2017, Payeur et al., 2021]. They show how different dendritic compartments can coordinate learning across layers. They still need feedback that gives useful information to each receiving cell.

Francioni et al. provided an experimental point of contact for this theoretical work [Francioni et al., 2026]. Their BCI task assigned opposite causal roles to nearby cortical neurons. After the effect of somatic activity was removed, the remaining dendritic signals reflected those roles and predicted later changes in activity. Interfering with the relevant dendritic process impaired learning. The study therefore found a dendritic signal that varied with the role of each neuron. It did not determine how the brain constructs this pattern. It also left open whether the measured activity teaches later synaptic change or contributes directly to the current computation.

2.2 Learning from scalar reward and local variation

A scalar reward can guide different cells in different directions when it is combined with another quantity that varies across neurons or synapses. REINFORCE formalized stochastic policy updates from a global performance signal [Williams, 1992]. Node perturbation estimates a useful direction by relating local exploratory changes to changes in an objective [Fiete and Seung, 2006]. Eligibility traces keep local information available until a later modulatory signal arrives [Izhikevich, 2007, Gerstner et al., 2018].

This principle has already been applied to BCI learning. A Hebbian model combined global reward with neuronal noise and reproduced selective network reorganization observed during a brain control task [Legenstein et al., 2010]. The local eligibility model used here belongs to this established family. We do not present node perturbation as a new algorithm. We use it to ask whether the kind of information missing from the pART model is enough to learn arbitrary cellular roles in the same abstract task.

2.3 Representation credit in ART

ART addresses the problem of learning new information without erasing stable knowledge. It does so through category matching, vigilance, search, and learned expectations carried by feedback [Carpenter and Grossberg, 1987]. pART extends these ideas to causal representations that must be kept active until a consequence arrives. A representation can be selected, preserved across a delay, and reinforced by a broad motivational signal [Grossberg, 2018]. These mechanisms explain how a population representation can receive credit. On their own, they do not specify an arbitrary feedback value for every neuron.

SMART provides a more detailed cellular circuit. In a model of the visual system, learned feedback reaches apical dendrites. A match between feedback and current activity changes the timing and participation of cells through a center and surround circuit. A local rule then changes synapses according to voltage and conductance [Grossberg and Versace, 2008]. Related laminar models describe similar forms of modulatory feedback [Grossberg, 2021]. Grossberg’s vector associative motor systems begin from a different source of information. They use a vector that already

describes the difference between the current and desired states [Gaudiano and Grossberg, 1991]. Adding such a vector here would assume the separate cellular instructions that the simulations are meant to explain.

The literature therefore suggests three different accounts. A feedback system may send a different error or target value to every cell. A population system may reinforce a useful pattern that is already present in a repertoire. A local exploratory system may combine one common outcome with a different eligibility trace at each cell. The experiments below ask whether population selection can reproduce the cellular signature usually associated with a vector of errors. When it cannot, the local trace provides one established source of the missing information.

3 Materials and methods

3.1 Task and available information

The experiments used abstract closed loop BCI tasks based on the information available in the Francioni paradigm. They were not detailed reconstructions of its cortical circuit. Each cell had a hidden causal role $c_i \in \{-1, +1\}$, and the two roles occurred equally often. For a somatic population pattern y , the environment calculated a signed drive from the difference between the mean activity of the P+ and P- groups. In the later standard tasks this was

$$u = \frac{2}{N} \sum_{i=1}^N (y_i - 0.5) c_i. \quad (1)$$

The drive changed a bounded visual state over several action frames in a closed loop. The learner then passed through irrelevant distractor steps before receiving a scalar measure of improvement and a binary success signal. Some studies used two observable contexts. In the second context the same cells had the opposite causal roles.

The learner could observe the context, visual state, its own activity, permitted local variables, and the delayed scalar outcome. It could not observe the hidden role vector c . That information was used only in labeled positive controls and offline analyses. We measured task performance, alignment between learned patterns and hidden roles, recovery after the roles were changed independently, and the effects of removing proposed causal mechanisms. An alignment of 1 indicates a perfect match with the hidden roles. We use *remapping* for an independently sampled change in those roles.

The models based on ART and SMART contain five quantities. The selected representation is $H_{k,h}$. Its learned feedback expectation is $T_{k,h,i}$. The modeled dendritic input is D_i , the plastic lower weight is W_i , and the somatic output is S_i . The main test was specified before the confirmatory run. It asked whether an early dendritic signal predicted a later change in weight, and whether that weight change predicted a later change in somatic output. The expected sequence was

$$D_{\text{residual},i}^{\text{early}} \longrightarrow \Delta W_i \longrightarrow \Delta S_i. \quad (2)$$

The test was evaluated while holding the selected hypothesis fixed. This prevents a switch between population patterns from appearing to be a change within individual cells. The dendritic measure was a *residual*, meaning the part of the modeled dendritic signal left after accounting for somatic activity.

The models based on SMART used the following qualitative plasticity rule.

$$\Delta W_{jk} \propto f_G(V_k, \bar{g}_{jk}) [\bar{g}_{jk} f_N(V_k) (\hat{w} - \check{w}) + w_0 - w_{jk}]. \quad (3)$$

In words, a synaptic weight changes according to conductance, voltage, spike timing, and the weight’s current value [Grossberg and Versace, 2008]. These are quantities available at or near the modeled synapse.

The generic local eligibility model used a different update. It is a simplified rule based on node perturbation, with three sources of information. It is not a new learning algorithm [Fiete and Seung, 2006, Izhikevich, 2007, Gerstner et al., 2018].

$$\Delta A_i = \eta_N (R - b) e_i. \quad (4)$$

Here, $R - b$ is the same scalar reward advantage for every cell. The eligibility trace e_i is a brief record of cell i ’s own exploratory change. Because the trace differs across cells, the same reward can produce different local updates. The factor η_N corrects the known scaling with population size. Hidden roles do not enter the update.

3.2 Common benchmark based on the Francioni task

The final benchmark gave the same information to every ordinary model. Ten cells were divided into five hidden P+ and five hidden P- roles. Their opposing activity moved a continuous visual state that was displayed in seven bins. A trial lasted at most 28 action frames of one second each. The learner observed the displayed state and its change during the trial, followed by a delayed binary reward. It never observed the hidden role vector. Initial learning and learning after an independent change of roles each lasted 14 blocks of 64 trials.

Five primary models were compared. The vector oracle received the hidden roles and served only as a check that the benchmark could detect the full signal. The ART learner selected from a fixed bank of 64 random balanced population patterns and learned feedback expectations. The eligibility learner used local perturbations together with visual and reward signals to adjust each cell. The combined model added that adjustment to the pattern selected by ART. The random model did not learn. Additional controls shuffled outcomes, removed exploration, removed the temporal trace, or disabled eligibility in the combined model.

Behavioral success alone was not taken as evidence of a dendritic teaching signal. Each model exposed its own feedback quantity $F_i(t)$ for every cell. The common observation layer generated a synthetic dendritic measurement

$$D_i(t) = \alpha S_i(t) + \beta \text{standardize}(F_i(t)) + \epsilon_i(t). \quad (5)$$

A separate linear relation between the soma and dendrite was then fitted for each cell, and the remaining signal was analyzed. The tests, fixed before the confirmatory run, covered behavior and topology. They also asked whether the dendritic signal had the correct direction for each role, carried information about the outcome, predicted later activity, and remained after matching somatic activity when enough comparable samples were available. The ART and eligibility components of the combined model were saved separately. This common observation layer is a measurement test, not a model of calcium biophysics.

3.3 Experimental sequence and reproducibility

The simulations were implemented in Python. Experiments with population activity used NumPy [Harris et al., 2020]. The reduced spiking and local plasticity model used Brian2 [Stimberg et al., 2019]. Randomness came from explicit seeds, and paired conditions shared the same hidden roles and initial repertoires. Later studies used separate seed sets that were not examined before confirmation. Their thresholds were also fixed in advance. Positive controls received the hidden role

vector, while other controls shuffled outcomes or removed exploration or eligibility. Table 1 links the scientific study names to the repository identifiers. The Results follow this order but use the scientific names in the text.

3.4 Statistical analysis

The hidden mapping and its paired initial conditions formed the independent unit in confirmatory comparisons. Unless otherwise noted, intervals from the earlier studies are 95% bootstrap intervals obtained by resampling those units 5,000 times. EXP006 reports 95% normal intervals across 32 mappings that were not examined before confirmation. Its pass or fail decision used averages and numerical thresholds committed before those mappings were accessed. Each control addressed a different possible reason for success or failure. Controls given the hidden vector showed that the task and the test of later prediction were sensitive when the correct directions for individual cells were available. Contextual bandits tested whether behavior required ART category dynamics. Shuffled outcomes tested whether local traces had to remain paired with the correct consequences. Removing exploration or eligibility tested the two local parts of the generic update. Exact replay checked whether the recorded permitted variables could reconstruct the learned quantities.

3.5 Scope of the claims

All quantitative results come from committed protocols, fixed analyses, records that were set aside for confirmation, or later audits that are clearly identified in the repository [OgnjenX, 2026]. EXP006 was designed after the earlier synthesis showed that the existing experiments did not fully answer the motivating question. Its protocol, parameters, source hashes, tests, and passing thresholds were fixed before the confirmatory run. The first study tests routing, not the learning of new cellular directions. Conclusions from models based on the theory apply to the models and tests implemented here, not to every ART, pART, or SMART model. The local eligibility model uses a generic rule with three sources of information and is not attributed to Grossberg. Biological interpretations remain hypotheses unless they are supported by the cited experiments.

3.6 Link to published theory

We compared every step of the model with the cited sources. pART motivated the selection and maintenance of representations. ART motivated expectation learning, matching, and search. SMART motivated apical feedback and local plasticity that depends on timing. The dendritic measure was based on Francioni et al. These sources did not explain how a scalar outcome alone could create an arbitrary pattern from H to T across individual neurons. They also did not establish two biological links used in the model. One connects the SMART surround to dendritic signals that depend on NDNF cells. The other connects modeled current or voltage to the calcium signal left after accounting for the soma. These gaps motivated the later tests of topology and local eligibility.

Registry ID	Study	Independent units	Purpose
EXP000	Routing a built in opponent pattern	30 seeds	Route an already available pattern.
EXP001	Selection from a random repertoire	30 mappings	Test selection from scalar outcomes after a delay.
EXP002	Learning a feedback expectation	30 mappings	Test whether an expectation supplies the later cellular signal.
EXP003	Comparison with published theory	Source audit	Identify supported links and unresolved mappings.
EXP003a	Reduced SMART motif	Mechanism tests	Isolate feedback, timing, and local plasticity.
EXP003b	Complete model	12 mappings	Test the complete sequence based on the theory.
EXP004	Origin of learned topology	16 mappings	Separate construction from inheritance of repertoire geometry.
EXP005	Local eligibility	16 mappings per N	Audit the published mechanisms, then test generic cellular credit.
EXP006	Common benchmark	32 mappings	Compare selection, cellular learning, and their direct combination in one fixed test.

Table 1: Crosswalk between repository identifiers and the scientific studies described in the manuscript.

4 Results

4.1 Routing a pattern already present

The first study, EXP000 in the repository, asked whether selective feedback could produce opposite effects in two groups of cells when the required pattern was already built into the model. After two retained calibration attempts, the corrected model reached an accuracy of 0.978 late in the task across 30 seeds. Modulation was +0.673 for P+ cells and −0.673 for P- cells (Figure 1). The difference between the groups was +1.346. Selective feedback could therefore route a strong opponent pattern without a separate error value for every neuron on each trial.

This showed that the proposed routing was possible, but it did not test learning at individual cells. P+ and P- identity could be recovered from even and odd neuron indices, and the same identity was encoded in the category templates. Freezing category and value learning did not reduce accuracy. Every ordinary corrected trial matched a stored category, and none triggered a new search (Table 2). The model could use a pattern that it already possessed. It did not explain how the pattern could be found when the roles of the cells were hidden.

4.2 Selecting from a random repertoire

The next study, EXP001, hid the roles of the cells and replaced the opponent template with a bank of random patterns. It asked whether a scalar outcome could select a useful pattern, preserve it across irrelevant events, and select another useful pattern after the hidden roles changed. The model succeeded at all three tasks.

Success depended strongly on what the random bank contained. Larger banks were more likely to include a pattern that already aligned well with the hidden roles, and the best initial alignment

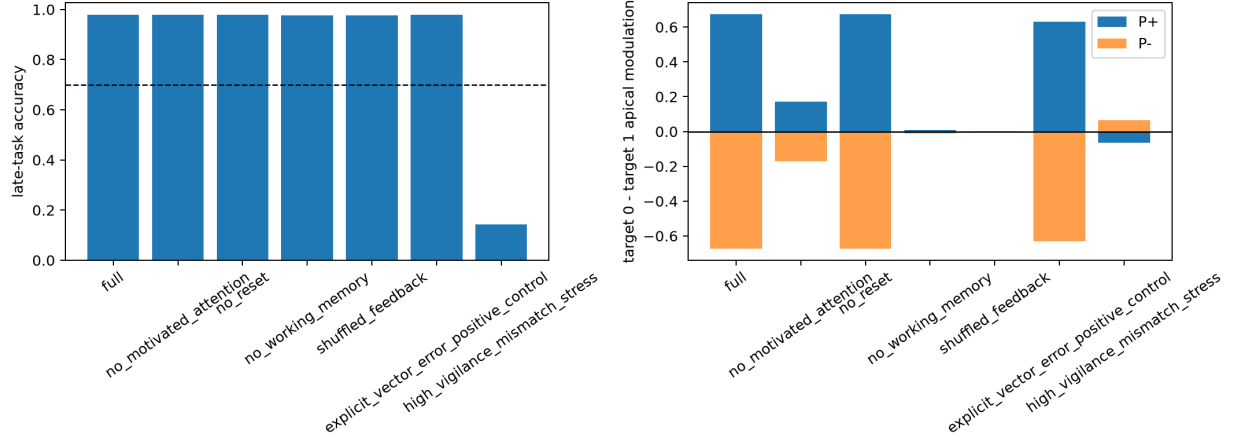


Figure 1: Summary of routing when the opponent pattern is already built into the model. Accuracy late in the task remains high after several mechanisms are removed because the required pattern is already present. Shuffled feedback disrupts P+/P- modulation. The explicit vector provides a positive control.

increased with bank capacity (Figure 2). This result places the achievement at the representation level. Reinforcement selected a useful direction, but that direction had to be present before learning began.

The remaining dendritic signal did not solve the problem of cellular credit. Its immediate alignment with the hidden roles also appeared in controls that did not learn. More importantly, it did not reliably predict how the activity of each cell would change later. A positive control with a separate teaching value for every neuron did predict that change. The next study asked whether a learned feedback expectation could generate the missing signal without receiving the hidden role vector.

4.3 Learning a feedback expectation

The next study, EXP002, asked whether repeated experience with selected patterns could produce a useful feedback signal. It added an association based on the Grossberg outstar.

$$T_h \leftarrow T_h + \eta(S_{\sim} - T_h). \quad (6)$$

The equation moves the expectation T_h toward the somatic pattern $S_{\sim}$ that repeatedly occurs with the selected hypothesis. The model learned this expectation and learned it again after the roles changed. Table 2 reports the behavioral scores and the alignment of the feedback pattern.

These behavioral results did not show that the learned expectation carried a teaching signal for individual cells. Simply copying the selected pattern produced similar alignment, and a contextual bandit matched the behavioral performance. The main test within each cell also failed. Before and after the roles changed, the remaining dendritic effects were -0.326 and -0.016 . The corresponding effects in the positive control, which received one teaching value for each neuron, were 0.660 and 0.545 . Suppressing expectation learning removed the alignment of the learned pattern but had little effect on behavior.

The model could store an expectation, but the expectation did not yet explain later changes in individual cells. The next step was to test whether matched feedback could control local plasticity.

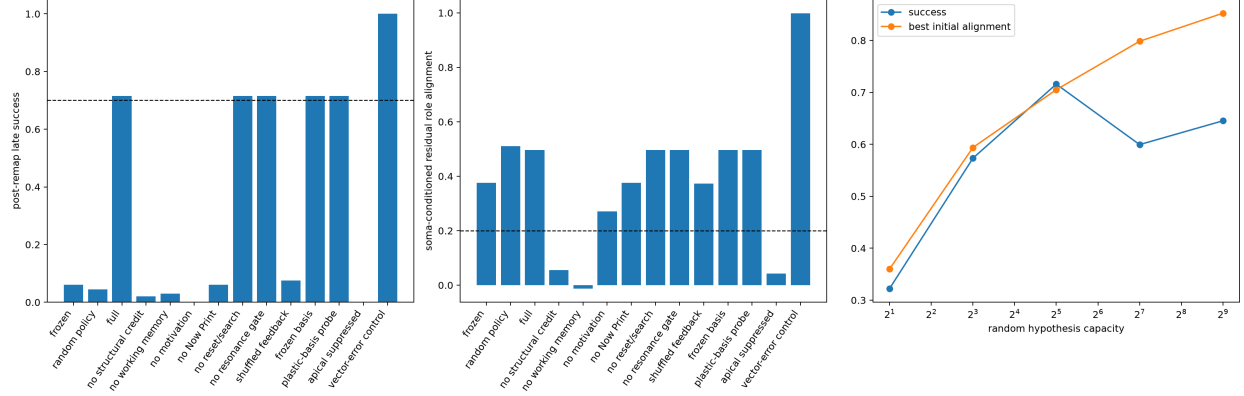


Figure 2: Comparison using a fixed random repertoire. The left panel shows success after the roles changed. The center panel shows how the remaining dendritic signal aligns with the roles after accounting for somatic activity. The right panel relates success to the best pattern available in the initial bank. The result is consistent with selection from a fixed repertoire rather than the learning of new cellular directions.

4.4 Testing local plasticity and the complete model

The local mechanism study began with a comparison to the published theory, EXP003, and then tested a reduced circuit inspired by SMART, EXP003a. It asked a narrow question. Could supplied matching feedback alter spike timing, allow a local weight update, and change a later response to incoming activity? The model did all three. Table 2 reports the update and spike results. No update occurred when a competing pattern was present, when feedback was removed, or when feedback and activity did not match. This established the proposed local sequence when the appropriate feedback was supplied. It did not explain how the system learned that feedback.

The complete model, EXP003b, combined this local mechanism with delayed representation selection and learned expectations. Across 12 mappings set aside for confirmation, feedback made local spikes occur earlier and changed lower weights. Removing SMART input, the learned expectation, or local learning eliminated the effect. Table 2 reports the timing result.

However, the model failed its main test, which had been specified in advance. The early dendritic signal did not predict the later change in weight or the later change at the soma well enough to meet the fixed threshold. The positive control with one value for every neuron did complete this chain of predictions. A later analysis asked where the information was lost. The learned expectation T and variables used directly by the plasticity rule predicted the sparse weight changes. The dendritic signal left after accounting for the soma did not. The correlation from raw T to ΔW was 0.416 [0.290, 0.530]. The correlation from the remaining D signal to ΔW was -0.004 $[-0.141, 0.128]$. This later analysis did not change the failed result. It raised two questions. Where did the pattern in the expectation come from? Did the residual measure preserve the relevant information?

4.5 Tracing the learned neuron pattern

The topology study, EXP004, asked whether the complete model constructed a new pattern across neurons or inherited one from its fixed repertoire. The test varied the quality of the best pattern available before learning. Better initial coverage produced better behavior and better alignment between the learned expectation and the hidden roles (Figure 3). With a bank size of $M = 16$, the difference in behavior between high and low coverage was 0.197 [0.137, 0.255]. Table 2 reports the

corresponding scores and alignments.

A constructed sequence showed that the model could combine behavior across steps. Its performance reached 0.980 and exceeded the best single action repeated across steps by 0.313 [0.260, 0.374]. However, a simpler contextual bandit reached 0.977. Combining actions over time therefore did not show that the model had constructed a new pattern across cells.

The source of the learned pattern became clear through a comparison with experienced activity.

$$\text{corr}(T, \text{selected initial soma}) = 0.999851, \quad \text{corr}(T, \text{mean target}) = 0.999870. \quad (7)$$

Exact replay reconstructed T with RMSE 0.0, and T did not surpass the best individual target that the model had experienced. Permuting outcomes changed which hypotheses and targets were visited. This explained most of the change in topology. The outstar learned from selected states, but the detailed pattern across neurons came mainly from the fixed repertoire. Learning an arbitrary new topology would require some information that varies across cells and is not already present in a candidate pattern.

4.6 Testing a local eligibility trace

The next study, EXP005, first asked whether a published Grossberg equation supplied the missing information for each cell. pART, ART and outstar learning, exploration, SMART, and vector motor systems each provided a related component. None of the sources we inspected combined an independent local trace of exploration with a delayed scalar outcome to learn arbitrary signs in intermixed cells. We therefore did not construct another model and call it a Grossberg model.

Instead, a separate generic model tested whether this additional operation was enough. After each exploratory change, every cell retained a private eligibility trace e_i . A later scalar outcome strengthened changes that had helped and reversed changes that had hurt. The update never received the hidden roles.

The model learned the hidden signs at all four population sizes and learned them again after an independent change of roles (Table 3). At $N = 32$, alignment reached 0.987 during initial learning and 0.964 after the roles changed. The learning curve rose quickly, fell when the hidden map changed, and rose again as the new map was learned (Figure 4). Shuffling the outcome reduced the final alignment to 0.229. Removing plasticity, the temporal trace, or exploration left alignment near zero. The difference between the ordinary and shuffled conditions was 0.735 [0.655, 0.820].

Exact replay reproduced the updates to numerical precision. At every population size, the final topology was also better than the best single exploratory sample. The model therefore accumulated information across trials instead of selecting one fortunate perturbation. The scalar reward carried no more information than before. The local trace supplied the missing information by allowing the same reward to produce different updates at different cells.

4.7 Comparing the mechanisms in one common benchmark

The final study, EXP006, returned to the motivating question using one task, one observation model, and one classification rule fixed in advance. It compared the vector oracle, the ART repertoire learner, the local eligibility learner, their direct combination, and a random control. All were tested on 32 previously untouched mappings with $N = 10$.

Only the vector oracle passed the complete benchmark (Table 4). This shows that the task and analysis could detect every required result when the correct direction for each cell was supplied. It does not make the oracle biologically plausible.

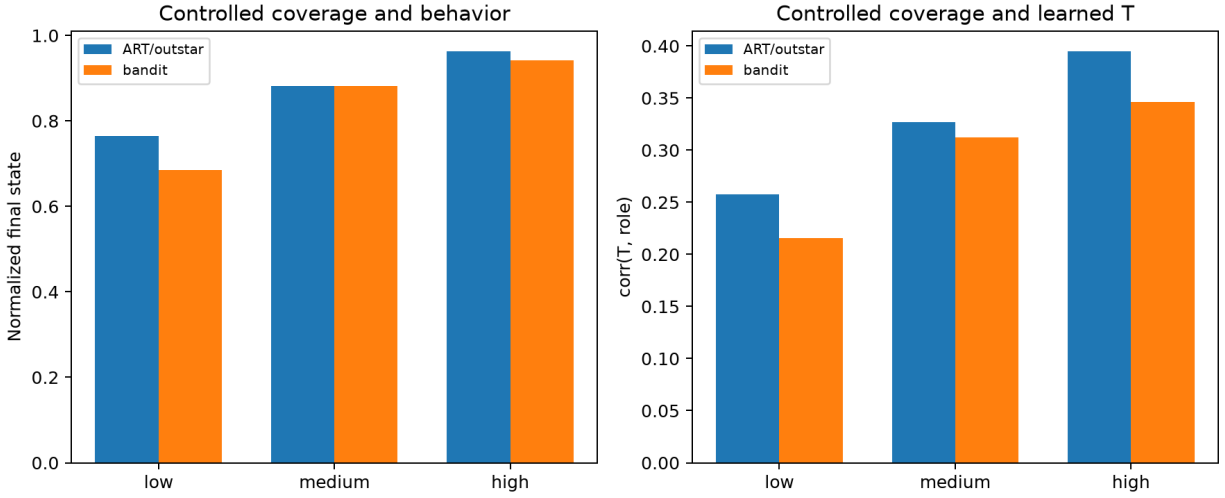


Figure 3: Result of the controlled coverage test. Final behavior on the left and alignment of the learned feedback expectation T on the right both improve when better patterns are already available in the bank. The contextual bandit follows the same trend.

Study	Supporting comparison	Result
Routing	Accuracy with category and value learning frozen	0.978
Routing	Matches with stored categories on ordinary corrected trials	100%, no resets
Learned expectation	Behavior before and after remapping	0.528, 0.397
Learned expectation	Feedback alignment before and after the roles changed	0.343, 0.287
Local plasticity	Modeled update and spike test with learning disabled	0.600 $\rightarrow$ 0.731 and 0 $\rightarrow$ 7/8
Complete model	Earlier local spikes caused by feedback	6.29 ms
Topology	Behavior at low, medium, and high initial coverage	0.765, 0.882, 0.962
Topology	Learned- T alignment at the same coverage levels	0.258, 0.327, 0.395

Table 2: Supporting quantitative results from the studies based on ART and SMART.

N	Initial alignment	Alignment after role change	Sign reversal in changed cells
8	0.997	0.968	1.000
16	0.993	0.960	1.000
32	0.987	0.964	1.000
64	0.972	0.950	0.975

Table 3: Results for local eligibility across population sizes. Each value summarizes 16 independent maps after 1,280 episodes of initial learning and 2,560 episodes after the roles changed.

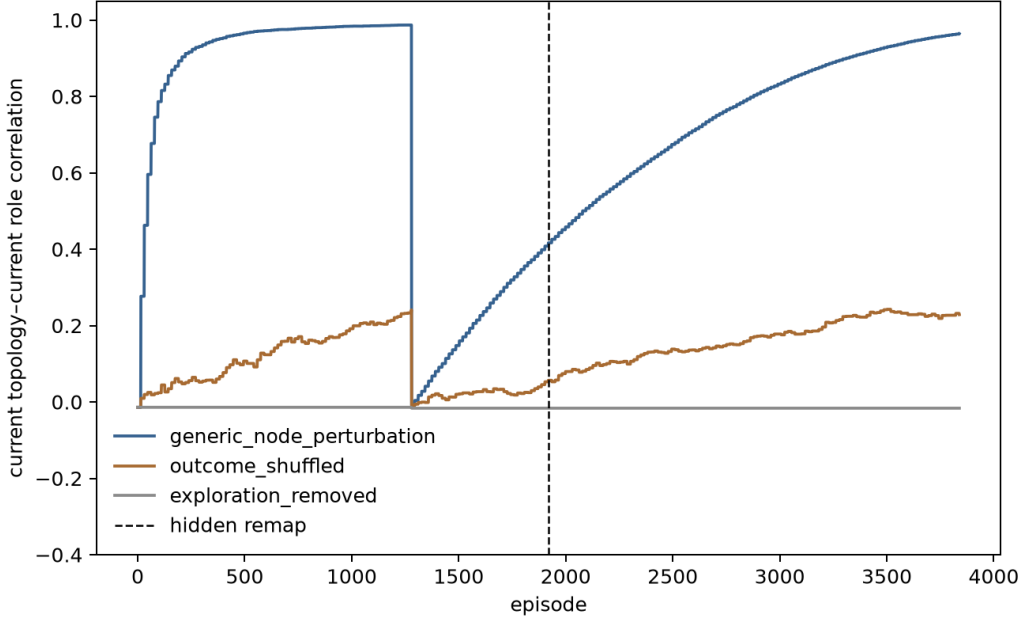


Figure 4: Learning with local eligibility at $N = 32$. Alignment learned from scalar outcomes and local perturbations rises during initial learning. It drops when the hidden map changes and then recovers as the new map is learned. Shuffled outcomes produce only weak alignment, while removing exploration prevents learning.

The ART learner passed the tests of behavior and topology. Its success late in training was 0.675 during initial learning and 0.744 after the roles changed. Its topology correlations were 0.613 and 0.663. The internal signal scores for the interaction between cell role and movement direction were 0.513 and 0.613. The corresponding scores in the measured residual were 0.030 and 0.039. This residual effect remained positive after matching somatic activity, and the signal carried enough information to decode the outcome. However, the early residual did not predict the later change in activity during initial learning. That score was -0.203 $[-0.396, -0.010]$. ART therefore failed the fixed criteria based on the Francioni signal. Its final topology was also worse than the best pattern already available in its bank. This supports learned selection, not the creation of new cellular directions.

The eligibility learner reached a success of 1.000 during initial learning and 0.656 after the roles changed. It constructed a strong initial topology with a correlation of 0.985. After the change, the correlation reached only 0.489, and the fraction of changed cells that reversed sign reached 0.673. Both values fell below their fixed thresholds. The internal and measured signals for role and movement direction were near zero, and outcome decoding remained close to chance. The learner therefore showed substantial adaptation in individual neurons without reproducing the dendritic signature.

The combined model passed behavior and topology. Its success was 1.000 during initial learning and 0.905 after the roles changed. Its topology correlations were 0.878 and 0.763. The dendritic criteria still failed. The combined internal signal scores were -0.119 and -0.088 . The measured residual scores were -0.018 and -0.013 . Saving the two components separately revealed the source of this conflict. The ART component was positive at 0.555 and 0.591. The eligibility component was negative at -0.127 and -0.096 . Combining two operations that both helped behavior did not automatically produce the signal observed by Francioni et al. (Figure 5).

All five control comparisons passed. Shuffling outcomes abolished ART success and strongly reduced eligibility learning. Removing exploration or the temporal trace also abolished eligibility learning. Enabling eligibility in the combined model improved initial success by 0.323 and initial topology by 0.278 compared with the same model without eligibility plasticity. The two saved components reconstructed the combined signal exactly in every condition.

The planned $N = 8$ check retained the main pattern among the ordinary models. Every ordinary learner again failed the complete dendritic criterion. This was not a clean replication of the full classification because even the vector oracle missed the threshold for decoding before the outcome. The smaller task also favored repertoire coverage. A bank of 64 patterns contains nearly all 70 possible balanced patterns for eight cells. The $N = 8$ result is therefore a qualitative check, not a second primary test.

Model	Behavior	Topology	Dendritic signature	Full bridge
Vector oracle	Pass	Pass	Pass	Pass
ART repertoire	Pass	Pass	Fail	Fail
Local eligibility	Pass	Fail	Fail	Fail
ART plus eligibility	Pass	Pass	Fail	Fail
Random no learning	Fail	Fail	Fail	Fail

Table 4: EXP006 classification for the primary $N = 10$ benchmark. The dendritic column combines four tests fixed before confirmation. They cover the internal signal, the measured residual, information about the outcome, and prediction of later activity.

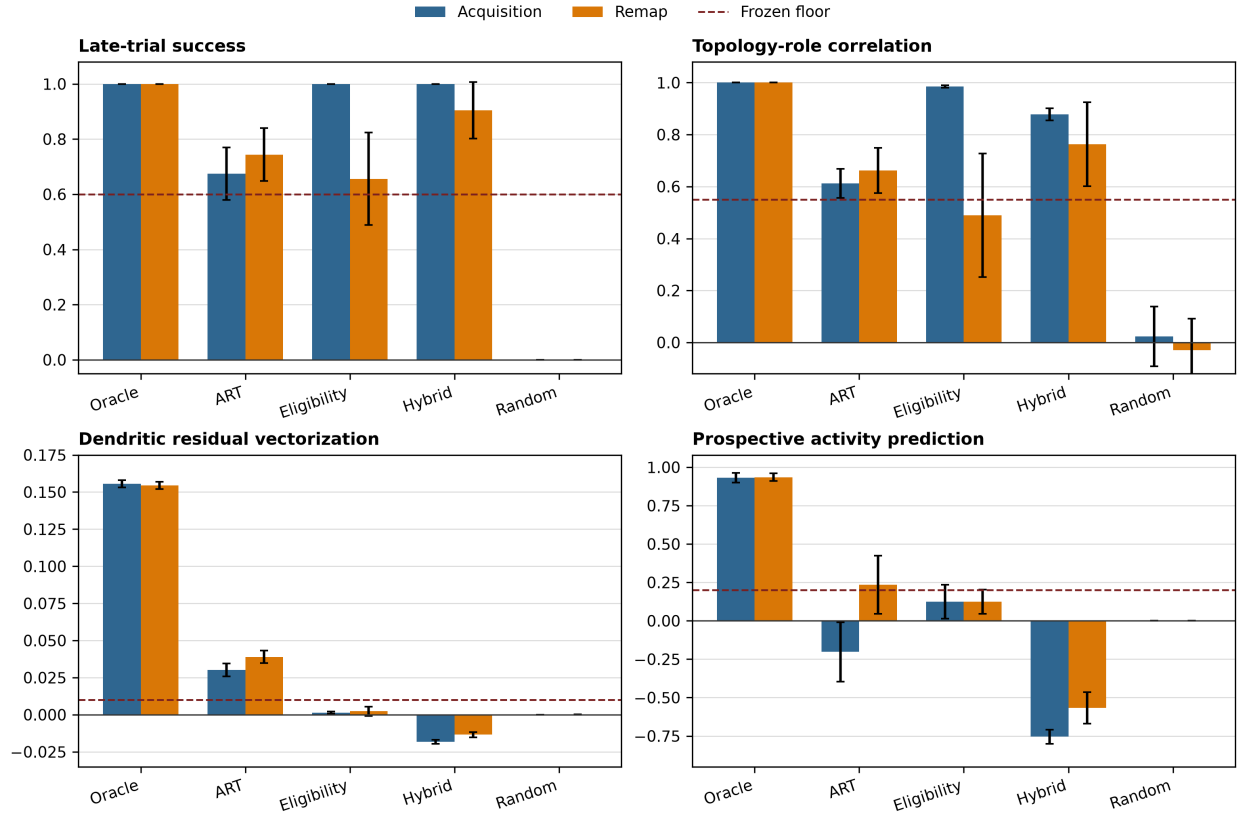


Figure 5: Primary result from EXP006. ART, local eligibility, and their combination all improve behavior, but they differ in topology and in the dendritic tests. Dashed lines mark the thresholds fixed before confirmation. Error bars show 95% intervals across 32 independent mappings.

5 Discussion

5.1 What the study shows

The experiments separate two kinds of learning that can produce similar behavior. Scalar reinforcement was enough to select and maintain useful population patterns. It also supported learned expectations and learning after the roles changed. However, the detailed structure across neurons in the ART models came mainly from patterns that were already available. Selecting a successful representation is not the same as learning how each cell should change.

The common benchmark shows that this distinction is not simply caused by using different tasks. With the same feedback and evaluation, ART improved behavior by selecting patterns from its bank that aligned with the hidden roles. Local eligibility learned a new topology during initial training. Their combination learned fastest and did both, but it failed the complete dendritic criterion. A model can therefore solve behavior and topology without reproducing the measured cellular signal.

5.2 Do error based learning and ART describe the same process?

The original question was whether these two accounts could describe the same process. The answer depends on the level being considered. At the level of behavior and population selection, they can look similar. Both a learner driven by error and a system that selects representations can improve performance when the task changes. At the level of individual cells, however, they were not interchangeable in the models tested here. An error or target for each role already contains information that differs across cells. A representation selected by pART identifies which distributed state deserves credit, but it does not automatically create the detailed pattern within that state.

This does not make the two accounts incompatible. They may describe different parts of one learning system. An ART mechanism could stabilize context, select a causal representation, and preserve it until an outcome arrives. A separate cellular mechanism could then determine how eligible synapses or neurons change. That mechanism might use a vector of instructions, a local perturbation trace, a dendritic target, or some combination of these signals. The experiments show that the particular bridge from pART tested here is not equivalent to a cellular error mechanism. They also show that one direct combination is insufficient, but they do not rule out other ways of joining the mechanisms.

The failed tests of later change are important for the same reason. The earlier complete model learned expectations, changed spike timing, and changed local weights. Even so, its dendritic signal after accounting for the soma did not predict later change. In EXP006 the combined model learned behavior and topology, but its ART and eligibility components contributed opposite signs to the common dendritic measure. These failures do not contradict the Francioni experiment or disprove ART or SMART. They show that joining individually useful components is not enough. A model must explain how the correct information for each cell is created, combined, and preserved in the measured dendritic signal.

5.3 Relation to previous findings

The eligibility result agrees with established work on perturbation learning. A scalar reward can guide different cells when it is combined with a local quantity that differs across cells. Dynamic node perturbation can estimate a useful direction from changes in a global objective [Fiete and Seung, 2006]. Eligibility traces can preserve local information until a later modulatory event [Izhikevich, 2007, Gerstner et al., 2018]. An earlier BCI model also reproduced selective network change by

combining global reward with neuronal noise in a Hebbian rule [Legenstein et al., 2010]. Our eligibility model applies this known principle to the information that was missing from the ART models.

The comparison with Francioni et al. gives a more limited result [Francioni et al., 2026]. ART generated a signal related to cell role, and the learning models improved behavior, so they reproduced parts of the abstract task. However, ART did not create a new pattern across cells, and eligibility did not reproduce the measured dendritic signal. Their direct combination also failed to reproduce the early dendritic signal that predicts later cellular change. This does not conflict with the biological finding. It identifies the part of that finding that the tested models do not yet explain.

The negative result also does not reject ART or SMART. Those theories motivated population selection, learned expectations, matching, timing, and local plasticity. They did not supply a complete mechanism for creating arbitrary instructions for individual neurons from a scalar outcome. The present results therefore agree with earlier work on scalar reward and eligibility while placing a clear limit on the ART bridge tested here.

5.4 Measurement and attribution boundaries

The complete models also revealed a problem of measurement. Information can be present in a learned expectation, an eligibility signal, a current, or spike timing and still be absent from the remaining calcium signal. EXP006 used one common observation layer and saved the internal signal separately from the measured residual. This helped distinguish failure of the learning rule from loss of information in the measurement. It did not make the synthetic observation layer biological. Future studies should state in advance which variable is expected to cause learning and how it should appear in dendritic recordings.

The comparison with published theory also sets an important boundary. A model may resemble pART or SMART without every component following from those theories. The cited work motivates representation selection, learned expectations, matching, and local plasticity. The exploratory eligibility rule adds a new hypothesis. It should be tested as a separate mechanism rather than attributed to Grossberg.

The biological interpretation also remains open. Francioni’s signal could be an instruction for each neuron, an error computed locally, the effect of matching feedback, eligibility for later reinforcement, or another dendritic operation. The simulations compare the information required by these accounts. They do not identify the biological signal with any one of them.

6 Limitations

These tasks are abstract simulations. They do not reconstruct retrosplenial circuitry, animal behavior, calcium dynamics, or every equation in pART and SMART. Several biological links remain assumptions. They include a connection from a representation selected by pART to a projection resembling SMART, the targeting of relevant BCI cells, the mapping from center and surround organization to NDNF circuitry, and the conversion of modeled activity into the remaining calcium signal. The first routing model also begins with a favorable structure. Results for the random repertoire and outstar depend on the pattern banks and assignment rules that were tested.

The common dendritic layer deliberately maps each model’s internal value for every cell into the same synthetic observation. A positive result shows that the value can survive this chosen transformation. It does not show that real dendrites use that value or that real calcium follows the same equation. A negative result can arise from the learning mechanism, the chosen internal

variable, or the way it enters the observation layer. The successful oracle and the separately saved internal signals make these possibilities easier to see, but do not eliminate them.

The complete model used sparse local plasticity with discrete changes. Later analyses help explain why its main test failed, but they do not change the original result. The local eligibility model includes a correction for population size, but no hidden role information. Even so, success in a finite and normalized task does not show that the rule would scale in the brain. EXP006 tested one ART implementation, one eligibility rule, and one direct combination. Its failure on the complete benchmark does not apply to every ART model or every possible combination.

The secondary $N = 8$ benchmark could not support the full test because the vector oracle missed one decoding threshold. Its repertoire also covered nearly all possible balanced patterns, making it harder to distinguish selection from the construction of a new pattern. The $N = 10$ result is therefore primary. Finally, this manuscript brings together computational records from the repository. It is not an independent reproduction or peer review of the simulations or of the motivating animal experiment.

7 Biological predictions and discriminating tests

The competing explanations make different experimental predictions.

1. **Interaction between eligibility and outcome.** A deviation in one cell before the outcome should leave a local trace. Combining that trace with the later scalar outcome should predict the direction of plasticity. Disrupting the interval between the deviation and the outcome should impair learning without eliminating the response to reward.
2. **Remapping within the same cell.** If selected feedback determines the expressed pattern, a change in context or reward contingency should reorganize feedback before later somatic activity changes. The same cell should also be able to reverse its role across contexts.
3. **Separation of selection and cellular adaptation.** Restricting the available population repertoire should impair selection by ART. Reducing variation between individual cells while preserving the population state should impair learning based on eligibility. A combined system should show both effects.
4. **Measurement of center and surround.** Excitatory feedback and the inhibitory surround associated with NDNF cells should be measured and perturbed independently. This would test whether the predictive signal is a signed effect on the membrane rather than an excitatory calcium signal alone.
5. **Intervention at a specific projection and time.** A candidate source of feedback should be perturbed before the outcome while sensory error and reward are preserved. This would show whether that projection carries information for individual cells. Broad suppression of activity would be harder to interpret.

Together, these tests could distinguish among separate instructions for each cell, selection of an available population pattern, and credit calculated from a local trace of exploration.

8 Conclusion

This study began by asking whether cellular learning based on error and population learning based on ART might explain the same process. The experiments do not support a direct equivalence. A scalar outcome was enough to select and maintain useful population patterns without a separate teaching value for every neuron. In the pART models tested here, however, the detailed structure across neurons came mainly from representations that were already available. The earlier complete

model also failed the test, specified in advance, that linked an early dendritic signal to later changes at the synapse and soma.

A separate generic rule learned arbitrary cellular roles by combining the scalar outcome with a local eligibility trace. The common benchmark then tested whether the two operations together explained the motivating result. ART improved behavior by selecting useful patterns that already existed. Eligibility produced substantial adaptation in individual cells. Their direct combination passed the tests of behavior and topology, but its combined dendritic signal had the wrong sign. Only the vector oracle passed every criterion fixed before confirmation.

The central conclusion is that the two forms of learning came apart under the same test. Selecting a causal representation and assigning credit to its individual cells are different operations. They may coexist in one biological system, but one does not automatically produce the other. Simply adding them is not yet an explanation of the Francioni signal. The most direct next step is to compare candidate mechanisms using the same biological measurements. A decisive experiment would test whether a spontaneous or imposed change in one cell leaves a local trace before the outcome. The key question is whether combining that trace with later reward predicts the direction of plasticity. In parallel, a more detailed biological model should test how feedback, inhibition, and local eligibility combine, and whether the resulting signal survives the calcium analysis used in the motivating experiment after somatic activity is removed.

Data and code availability

All computational results reported here come from archived protocols, fixed analyses, experiment records set aside for confirmation, and later audits that are clearly labeled. The EXP006 protocol, parameters, source hashes, tests, and numerical thresholds were committed before the confirmatory run. The public project repository provides the source code, experimental records, analysis outputs, and figures used in this manuscript.

References

- Gail A. Carpenter and Stephen Grossberg. ART 2: Self-organization of stable category recognition codes for analog input patterns. *Applied Optics*, 26(23):4919–4930, 1987. doi: 10.1364/AO.26.004919.
- Ila R. Fiete and H. Sebastian Seung. Gradient learning in spiking neural networks by dynamic perturbation of conductances. *Physical Review Letters*, 97:048104, 2006. doi: 10.1103/PhysRevLett.97.048104.
- Valerio Francioni, Vincent D. Tang, Enrique H. S. Toloza, Zilan Ding, Norma J. Brown, and Mark T. Harnett. Vectorized instructive signals in cortical dendrites. *Nature*, 652:1254–1263, 2026. doi: 10.1038/s41586-026-10190-7.
- Paolo Gaudiano and Stephen Grossberg. Vector associative maps: Unsupervised real-time error-based learning and control of movement trajectories. *Neural Networks*, 4(1):147–183, 1991. doi: 10.1016/0893-6080(91)90002-M.
- Wulfram Gerstner, Marco Lehmann, Vasiliki Liakoni, Dane Corneil, and Johanni Brea. Eligibility traces and plasticity on behavioral time scales: Experimental support of neohebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12:53, 2018. doi: 10.3389/fncir.2018.00053.
- Stephen Grossberg. Desirability, availability, credit assignment, category learning, and attention. *Brain and Neuroscience Advances*, 2, 2018. doi: 10.1177/2398212818772179.
- Stephen Grossberg. A canonical laminar neocortical circuit whose bottom-up, horizontal, and top-down pathways control attention, learning, and prediction. *Frontiers in Systems Neuroscience*, 15:650263, 2021. doi: 10.3389/fnsys.2021.650263.

- Stephen Grossberg and Massimiliano Versace. Spikes, synchrony, and attentive learning by laminar thalamo-cortical circuits. *Brain Research*, 1218:278–312, 2008. doi: 10.1016/j.brainres.2008.04.024.
- Jordan Guerguiev, Timothy P. Lillicrap, and Blake A. Richards. Towards deep learning with segregated dendrites. *eLife*, 6:e22901, 2017. doi: 10.7554/eLife.22901.
- Charles R. Harris et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- Eugene M. Izhikevich. Solving the distal reward problem through linkage of STDP and dopamine signaling. *Cerebral Cortex*, 17(10):2443–2452, 2007. doi: 10.1093/cercor/bhl152.
- Robert Legenstein, Steven M. Chase, Andrew B. Schwartz, and Wolfgang Maass. A reward-modulated hebbian learning rule can explain experimentally observed network reorganization in a brain control task. *Journal of Neuroscience*, 30(25):8400–8410, 2010. doi: 10.1523/JNEUROSCI.4284-09.2010.
- OgnjenX. part-vector-credit: part-inspired credit assignment versus dendritic vectorization. GitHub repository, 2026. URL <https://github.com/OgnjenX/part-vector-credit>. Repository experimental record inspected 15 September 2026.
- Alexandre Payeur, Jordan Guerguiev, Friedemann Zenke, Blake A. Richards, and Richard Naud. Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24: 1010–1019, 2021. doi: 10.1038/s41593-021-00857-x.
- Marcel Stimberg, Romain Brette, and Dan F. M. Goodman. Brian 2, an intuitive and efficient neural simulator. *eLife*, 8:e47314, 2019. doi: 10.7554/eLife.47314.
- Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8:229–256, 1992. doi: 10.1007/BF00992696.